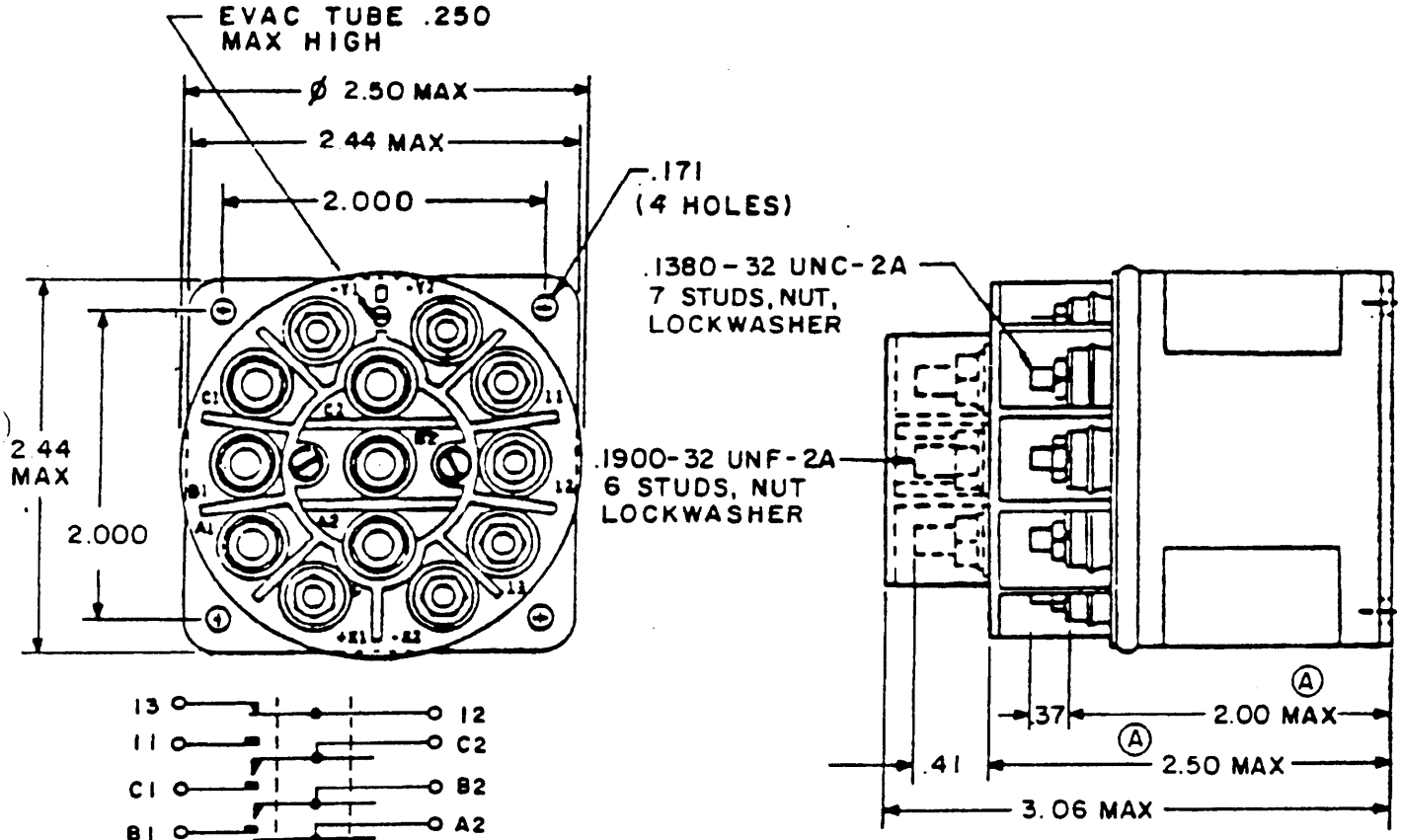
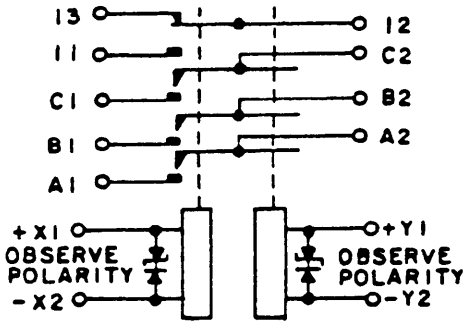
MILITARY SPECIFICATION SHEET

Ⓐ RELAYS, ELECTROMAGNETIC, 50 AMPERES, 115/200 V AC, 400 HZ, 3PST, WITH 5 AMPERE SPDT AUXILIARY, MAGNETIC LATCHING, HERMETICALLY SEALED

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-6106.





CIRCUIT DIAGRAM

INCHES	MM
.171	4.34
.250	6.35
.37	9.4
.41	10.4
1.95	49.5
2.000	50.80
2.44	62.0
2.48	63.0
2.50	63.5
3.06	77.7

FIGURE 1. Relay, outline drawing.

Ⓐ denotes changes

FSC 5945

AMSC N/A

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.030 (0.76 mm) for two place decimals and ±.010 (0.25 mm) for three place decimals.
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. Energizing +X1 and -X2 closes contacts (A1-A2, B1-B2, C1-C2); each remains closed after deenergized. Energizing +Y1 and -Y2 opens contacts (A1-A2, B1-B2, C1-C2); each remains open after deenergized.
6. With one coil energized at 28 V dc, energizing opposite coil shall not override it and trip relay.
7. Deenergize +X1 and -X2, and energize +Y1 and -Y2 through 11 and 12 shall trip contacts to open position and self deenergize.

FIGURE 1. Relay outline drawing - Continued.

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REQUIREMENTS:

Coil data:

Duty cycle: Continuous.

Operating voltage nominal: 28 to 30 V dc.

Pickup voltage, maximum: 18 V dc at +71°C.

Current rating: 0.087 ampere maximum at 28 V dc and +25°C.

Reset voltage: 18 V dc maximum at +71°C.

Coil resistance: 344 ohms ±10% at +25°C.

Coil transient voltage: Shall be limited to ±50 volts.

Contact data:

Operate time: 35 milliseconds maximum at +25°C nominal voltage.

Reset time: 35 milliseconds maximum at +25°C nominal voltage.

Contact bounce: 3 milliseconds maximum at +25°C nominal voltage.

Main contacts X1-X2, 115/200 V ac, 400 Hz, 3 phase:

Current rating:

Resistive: 50 amperes.

Inductive: 50 amperes.

Motor: 30 amperes.

Overload current: 500 amperes.

Rupture current: 500 amperes.

Minimum current: 1 ampere at 28 V dc or 5 amperes at 115 V ac, 400 Hz.

Auxiliary contacts:

Configuration: SPDT.

Voltage: 28 V dc or 115 V ac, 400 Hz.

Resistive current: 5 amperes.

Physical data:

Strength of terminals: Applicable.

Base cover: Melamine fiberglass, MIL-P-15037.

Terminal barriers: Nylon 6/6, type 1, grade A, class 1, per LP-395, black.

Weight: 14 ounces maximum.

Finish: Instrument black.

Environmental data:

Ambient temperature range: -55°C to +71°C.

Altitude: 60,000 feet.

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consist of this specification and the latest issue of

Vibration (sinusoidal): .06 inch DA at 10-57 Hz, 10 G's at 57-2,000 Hz, 50 G's for 6-9 milliseconds.

Centrifugal acceleration: 15 G's.

Life:

Electrical: 50,000 cycles minimum at rated load.

Mechanical: 100,000 cycles.

Part number: M6106/45-001.

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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